INCH-POUND
MIL-P-23469/1B(SH)
19 February 1988
SUPERSEDING
MIL-P-23469/1A(SHIPS)
31 May 1967

MILITARY SPECIFICATION SHEET

Ⓑ

COLLAR, GROOVED PIN-RIVET, SWAGE-LOCKED (LOCKPIN); REGULAR HEIGHT, FLANGED AND LOW PROFILE, ALUMINUM ALLOY, CORROSION-RESISTANT AND CARBON STEELS

Ⓑ This specification sheet is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

Ⓑ The requirements for acquiring the collars described herein shall consist of this specification and the latest issue of MIL-P-23469.

Collars, swage-locking shall be furnished in the following types and classes. Collars manufactured to this specification are intended for use with pins manufactured in accordance with MIL-P-23469.

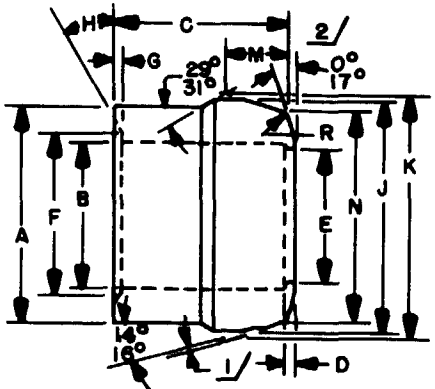
Type I - Regular height (see figures 1 and 2)
Type II - Flanged (see figures 3 and 4)
Type III - Low profile (see figures 5 and 6)
Type IV - Flanged (see figure 7)

Class 1 - Aluminum alloy
Class 2 - Corrosion-resistant steel
Class 3 - Carbon steel
Ⓑ Class 5 - Carbon steel high strength

Ⓑ Denotes changes.

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CLASS 1,2,3 AND 4 MATERIAL

SH 13203031

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M23469/1 6/																		
ALUM	CRES	Carbon steel	Carbon steel high strength		A	B	C	D	E	F	G	H	J	K	L	M	N	R
Class 1	Class 2	Class 3	Class 5	Size	Dia	Dia		Max	Min				3/	4/	5/		Ref dia	Max rad
-10106				3/16	.309 .306	.195 .197	.237 .247	.016	.190	.217 .227	.007 .011	15° 18°	.320 .314	.324 .320	.004	.105 .114	.264	
	-20106	-30106	-50106		.309 .306	.1895 .1925	.222 .232	.031	.186	.217 .227	.007 .011	15° 18°	.314 .310	.325 .321	.004	.102 .109	.267	.030
-10108				1/4	.406 .403	.261 .263	.308 .318	.016	.256	.285 .295	.009 .015	15° 18°	.423 .415	.428 .425	.004	.130 .139	.354	
	-20108	-30108	-50108		.406 .403	.260 .363	.294 .304	.093	.254	.285 .295	.009 .015	15° 18°	.420 .412	.424 .420	.004	.106 .120	.362	.035

See footnotes at end of table.

FIGURE 1. Type I - regular height collars for use with pins having six locking grooves.

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(B)

M23469/1 6/																		
ALUM	CRES	Carbon steel	Carbon steel high strength	Size	A Dia	B Dia	C	D Max	E Min	F	G	H	J 3/	K 4/	L 5/	M	N Ref dia	R Max rad
Class 1	Class 2	Class 3	Class 5															
-10110				5/16	.505 .501	.306 .308	.354 .364	.031	.301	.340 .350	.025 .031	28° 32°	.524 .516	.529 .526	.005	.159 .166	.440	
	-20110	-30110	-50110		.492 .487	.306 .310	.354 .364	.093	.301	.340 .350	.012 .018	15° 18°	.509 .501	.516 .512	.005	.114 .121	.451	.040
-10112				3/8	.608 .600	.3735 .3765	.431 .441	.031	.367	.418 .428	.031 .037	28° 32°	.624 .614	.631 .627	.006	.169 .176	.536	
	-20112	-30112	-50112		.598 .591	.377 .382	.433 .443	.093	.367	.441 .451	.012 .018	15° 18°	.610 .600	.615 .611	.006	.140 .147	.536	.045

Surface roughness RHR 125 maximum.

1/ 0.001 maximum protrusion permissible.

2/ Angle permissible but not a requirement - top of collar may be flat.

3/ Diameter.

4/ Theoretical intersection of 14/16 and 29/31 degree angles.

5/ "A" and "B" diameters to be concentric within 0.003 total indicator reading (TIR). 14/16 degree angle to be concentric with "A" diameter and "P" angle within "L" TIR.

(B) 6/ Part number example - M23469/1-10108

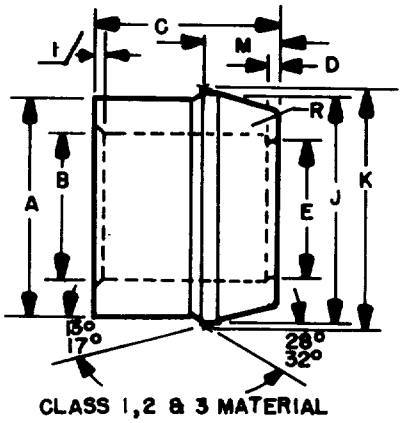
- Basic or general specification
- Collar
- Diameter in 1/32 inches (1/4)
- Style - type I (regular height)
- Material - aluminum alloy

NOTES:

1. MIL-P-23469/1B and MIL-P-23469/1A collars are interchangeable type for type, class for class, and size for size.
2. Dimensions are in inches.

FIGURE 1. Type I - regular height collars for use with pins having six locking grooves. - Continued

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SH 13203032

M23469/1 5/													
ALUM	CRES	Carbon steel		A	B	C	D	E	J	K	L	M	R
Class 1	Class 2	Class 3	Size	Dia	Dia		Max	Dia min	2/	3/	4/		Max rad
-10116	-20116	-30116	1/2	.795 .785	.520 .527	.615 .635	.125	.510	.842 .810	.853 .847	.008	.235 .250	.078
-10120	-20120		5/8	.979 .974	.651 .657	.798 .778	.165	.647	1.020 1.010	1.073 1.068	.008	.270 .300	.078
		-30120		.979 .974	.651 .657	.870 .850	.165	.647	1.020 1.010	1.073 1.068	.008	.270 .300	.078
-10124	-20124	-30124	3/4	1.176 1.170	.780 .787	.935 .955	.344	.776	1.220 1.210	1.275 1.269	.008	.344 .382	.078
		-30128	7/8	1.371 1.362	.910 .918	1.090 1.112	.375	.902	1.421 1.411	1.474 1.466	.008	.415 .455	.078
		-30132	1	1.564 1.556	1.039 1.047	1.245 1.272	.438	1.030	1.625 1.615	1.684 1.676	.008	.457 .507	.078

Surface roughness RHR 125 maximum.

1/ 1/32 x 45 degree maximum chamfer permissible at either end.

2/ Diameter.

3/ Theoretical intersection of 13/17 and 28/32 degree angles.

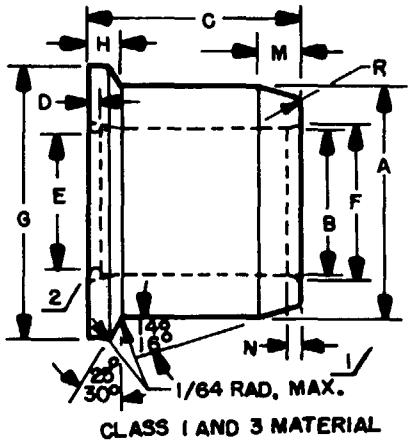
4/ "A" and "B" diameters to be concentric within 0.015 TIR. 13/17 degree angle to be concentric with "A" diameter within "L" TIR.

5/ Part number example - M23469/1-10116

- M23469/1 — Basic or general specification
- M23469/1 — Collar
- -10116 — Diameter in 1/32 inches (1/2)
- -10116 — Style - type I (regular height)
- -10116 — Material - aluminum alloy

FIGURE 2. Type I - regular height collars for use with pins having multiple locking grooves.

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SH 13203033

M23469/1 4/																
ALUM	CRES	Carbon steel	Carbon steel high strength	Size	A	B	C	D	E	F	G	H	L	M	N	R
Class 1	Class 2	Class 3	Class 5		Dia	Dia		Max	Min		Dia		3/			Max rad
-10206				3/16	.309 .306	.1895 .1925	.260 .270	.016	.186	.203	.385 .365	.042 .052	.004	.065 .080	.047	.030
	-20206	-30206	-50206		.311 .306	.1895 .1925	.260 .270	.016	.186	.203	.385 .365	.042 .052	.004	.065 .080	.047	
-10208				1/4	.406 .403	.261 .263	.359 .369	.016	.256	.281	.510 .490	.057 .067	.004	.095 .110	.062	.035
	-20208	-30208	-50208		.406 .403	.261 .263	.359 .369	.016	.256	.281	.510 .490	.057 .067	.004	.095 .110	.062	
-10210				5/16	.505 .501	.308 .310	.404 .414	.032	.301	.359	.635 .615	.073 .083	.005	.115 .130	.078	.040
	-20210	-30210	-50210		.505 .501	.308 .310	.404 .414	.032	.301	.359	.635 .615	.073 .083	.005	.115 .130	.078	

See footnotes at end of table.

FIGURE 3. Type II - flanged collars for use with pins having six locking grooves.

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M23469/1 4/																
ALUM	CRES	Carbon steel	Carbon steel high strength	Size	A Dia	B Dia	C	D Max	E Min	F	G Dia	H	L 3/	M	N	R Max rad
Class 1	Class 2	Class 3	Class 5													
-10212				3/8	.608 .600	.3735 .3765	.512 .522	.032	.367	.437	.781 .719	.089 .099	.006	.125 .140	.094	.045
	-20212	-30212	-50212		.608 .600	.3735 .3765	.512 .522	.032	.367	.437	.781 .719	.089 .099	.006	.125 .140	.094	

Surface roughness RHR 125 maximum.

1/ "F" maximum chamfer permissible for "N" maximum length.

2/ Inside diameter taper permitted for "H" depth, diameter not to exceed "B" plus 1/16 inch.

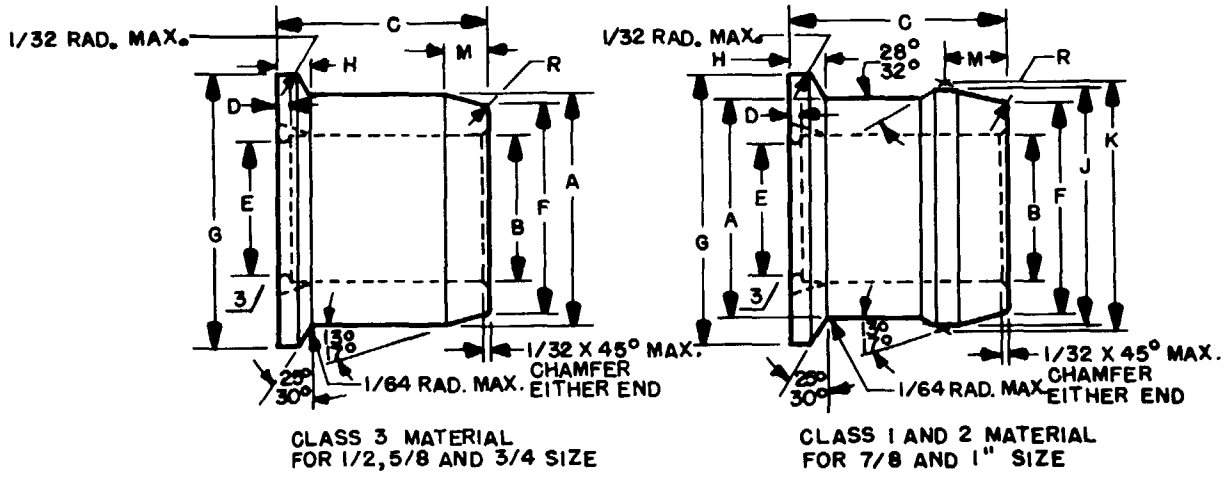
3/ "A" and "B" diameters to be concentric within .003 TIR. 14/16 degree angle to be concentric with "A" diameter and "P" angle within "L" TIR.

4/ Part number example - M23469/1-10208

- Basic or general specification
- Collar
- Diameter in 1/32 inch (1/4)
- Style - type II (flange)
- Material - aluminum alloy

FIGURE 3. Type II - flanged collars for use with pins having six locking grooves. - Continued

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SH 13203034

M23469/1 7/																	
ALUM	CRES	Carbon steel	Size	A	B	C	D	E	F	G	H	J	K	L	M	N	R
Class 1	Class 2	Class 3		Dia	Dia		Max	Min	Ref 1/			2/	4/	5/		6/	Max rad
	-20216		1/2	.805 .795	.520 .527	.720 .740	.125	.510	.700	1.031 .984	.120 .130			.006	.181 .191	.010	.031
-10216		-30216		.805 .795	.520 .527	.720 .740	.125	.510	.700	1.031 .984	.120 .130			.006	.181 .191	.010	.031
	-20220		5/8	.993 .983	.651 .657	.863 .883	.156	.647	.896	1.281 1.218	.151 .161			.008	.167 .177	.020	.047
-10220		-30220		.993 .983	.651 .657	.863 .883	.156	.647	.896	1.281 1.218	.151 .161			.008	.167 .177	.020	.047
	-20224		3/4	1.190 1.180	.781 .787	1.122 1.142	.188	.776	1.076	1.531 1.468	.183 .193			.008	.198 .208	.020	.062
-10224		-30224		1.190 1.180	.781 .787	1.122 1.142	.188	.776	1.076	1.531 1.468	.183 .193			.008	.198 .208	.020	.062

See footnotes at end of table.

FIGURE 4. Type II - flanged collars for use with pins having multiple locking grooves.

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M23469/1 7/																	
ALUM	CRES	Carbon steel		A	B	C	D	E	F	G	H	J	K	L	M	N	R
Class 1	Class 2	Class 3	Size	Dia	Dia		Max	Min	Ref 1/			2/	4/	5/		6/	Max rad
		-30228	7/8	1.371 1.364	.923 .930	1.305 1.325	.218	.915	1.264	1.625 1.562	.208 .228	1.426 1.416	1.501 1.494	.008	.415 .455	.031	.078
		-30232	1	1.564 1.556	1.034 1.042	1.490 1.510	.250	1.025	1.429	1.906 1.843	.240 .260	1.628 1.618	1.712 1.704	.008	.500 .540	.031	.078

Surface roughness RHR 125 maximum.

1/ Theoretical intersection.
2/ Diameter.
3/ Inside diameter taper permitted for "H" depth, diameter not to exceed "B" plus 1/16 inch.
4/ Theoretical intersection on 13/17 and 28/32 degree angles.
5/ "A" and "B" diameters to be concentric within 0.015 TIR. 13/17 degree angle to be concentric with "A" diameter within "L" TIR.
6/ "A" and "G" diameters to be concentric within "N" TIR.
7/ Part number example - M23469/1-10216

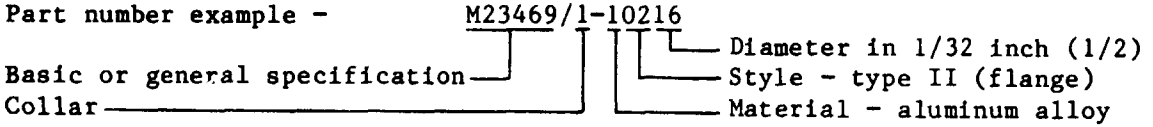
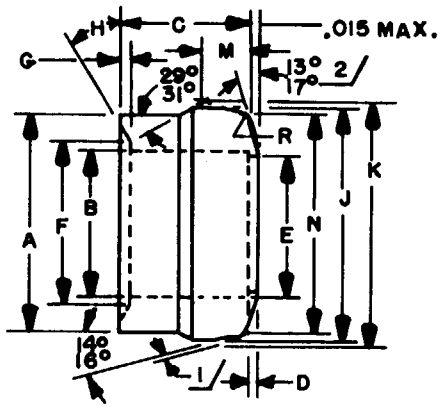


FIGURE 4. Type II - flanged collars for use with pins having multiple locking grooves. - Continued

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SH 12303035

M23469/1 6/																	
ALUM	CRES	Carbon steel		A	B	C	D	E	F	G	H	J	K	L	M	N	R
Class 1	Class 2	Class 3	Size	Dia	Dia		Max	Min				3/	4/	5/	Ref dia		Max rad
-10306	-20306	-30306	3/16	.309 .306	.1895 .1925	.142 .152	.016	.186	.217 .227	.007 .011	15° 18°	.314 .310	.325 .321	.004	.074 .078	.282	.030
-10308	-20308	-30308	1/4	.406 .403	.260 .263	.182 .192	.016	.254	.285 .295	.009 .015	15° 18°	.420 .412	.425 .419	.004	.097 .103	.368	.035
-10310	-20310	-30310	5/16	.492 .487	.304 .310	.235 .245	.031	.301	.340 .350	.025 .031	28° 32°	.509 .501	.516 .512	.005	.114 .121	.451	.040
-10312	-20312	-30312	3/8	.598 .591	.377 .382	.283 .293	.031	.367	.441 .451	.012 .018	15° 18°	.610 .600	.615 .611	.006	.140 .147	.536	.045

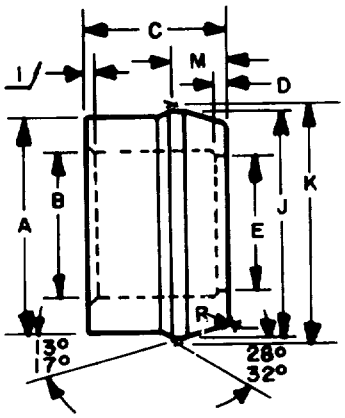
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FIGURE 5. Type III - low profile collars for use with pins having six locking grooves.

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Surface roughness RHR 125 maximum.
1/ 0.001 maximum protrusion permissible.
2/ 13/17 degree angle permissible but not a requirement - top of collar may be flat.
3/ Diameter.
4/ Theoretical intersection of 14/16 and 29/31 degree angles.
5/ "A" and "B" diameters to be concentric within 0.003 TIR. 14/16 degree angle to be concentric with "A" diameter and "P" angle within "L" TIR.
6/ Part number example - M23469/1-10308

Basic or general specification
Collar
Diameter in 1/32 inch (1/4)
Style - type III (low profile)
Material - aluminum alloy



CLASS 3 MATERIAL

SH 13203036

M23469/1 5/											
Carbon steel Class 3	Size	A Max	B Dia	C	D Max	E Dia min	J 2/	K 3/	L 4/	M	R Max rad
-30316	1/2	.785 .775	.527 .534	.475 .495	.125	.520	.815 .807	.859 .853	.008	.205 .220	.031
-30320	5/8	.979 .974	.651 .657	.675 .695	.165	.647	1.020 1.010	1.073 1.068	.008	.238 .253	.047
-30324	3/4	1.176 1.170	.781 .787	.720 .740	.188	.776	1.220 1.210	1.275 1.269	.008	.304 .319	.062
-30328	7/8	1.371 1.364	.923 .930	.838 .858	.218	.915	1.426 1.416	1.501 1.494	.008	.363 .383	.078

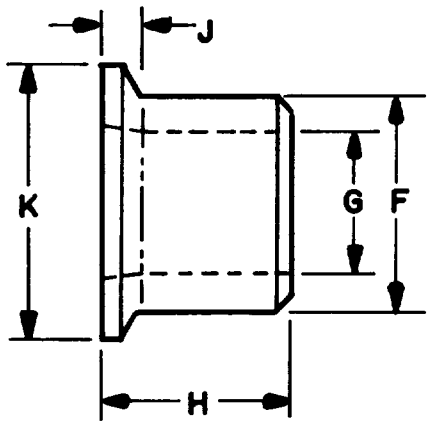
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FIGURE 6. Type III - low profile collars for use with pins having multiple locking grooves.

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Surface roughness RHR 125 maximum.
1/ 1/32 x 45 degrees maximum chamfer permissible at either end.
2/ Diameter.
3/ Theoretical intersection of 13/17 and 28/32 degree angles.
4/ "A" and "B" diameters to be concentric within 0.015 TIR. 13/17 degree angle to be concentric with "A" diameter within "L" TIR.
5/ Part number example - M23469/1-10316

Basic or general specification
Collar
Diameter in 1/32 inch (1/2)
Style - type III (low profile)
Material - aluminum alloy



SH 13203037

M23469/1 1/		Class 1 and 3 material					
ALUM Class 1	Carbon steel Class 3	Nom dia	F	G	H	J	K
10406	30406	3/16	.315 .308	.189 .195	.270 .240	.060 .032	.388 .362
10408	30408	1/4	.409 .402	.252 .258	.390 .360	.077 .047	.516 .488
10410	30410	5/16	.510 .502	.314 .321	.454 .422	.094 .061	.647 .602
10412	30412	3/8	.612 .604	.378 .385	.520 .490	.108 .078	.770 .730

See footnote at end of table.

FIGURE 7. Type IV - flanged collars for use with wide grip range pins having multiple locking grooves.

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M23469/1 1/	Class 1 and 3 material						
	Carbon steel Class 3	Nom dia	F	G	H	J	K
	30416	1/2	.797 .778	.512 .531	.677 .656	1/8	1
	30420	5/8	.995 .974	.641 .663	.847 .821	5/32	1-1/4
	30424	3/4	1.195 1.170	.777 .791	1.015 .985	3/16	1-1/2

1/ Part number example - M23469/1 - 10408

General specification
Collar
Diameter in 1/32 inch (1/4)
Style - type IV (flanged)
Material - aluminum alloy

FIGURE 7. Type IV - flanged collars for use with wide grip range pins having multiple locking grooves. - Continued

Preparing activity:
Navy - SH
(Project 5320-N024-1)

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